

Integrated Nav2 and MoveIt 2 Framework for Coordinated Mobile Manipulator Pick-and-Place Pipelines

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EXTENDED ABSTRACT

Seamless integration of 2D mobile base navigation with high-DOF arm manipulation remains a significant hurdle in modern robotics. This paper presents an integrated software pipeline bridging Nav2 (Navigation 2) costmaps with MoveIt 2 OMPL motion planners using ROS 2 action interfaces and synchronized TF2 coordinate trees. Evaluated on a TurtleBot3 Waffle Pi mounted with an OpenManipulator-X arm interacting with a fixed assembly arm, the system achieved sub-centimeter handoff positioning repeatability and reduced end-to-end cycle times by 28%.

Keywords: Nav2, MoveIt 2, Mobile Manipulation, TurtleBot3, OpenManipulator, Coordinate Frames, TF2

1. Coordinate Alignment & Software Bridging

Mobile manipulators require continuous coordinate transforms across multiple kinematic chains. Nav2 computes base velocities in /map -> /odom -> /base_link, while MoveIt 2 plans joint trajectories in /base_link -> /link1 -> /end_effector. The authors introduce a unified action pipeline that seamlessly triggers arm motion upon base docking completion without intermediate node restarts.

2. Kinematic Transform Formulations

The transformation from global map to gripper end-effector is expressed via homogeneous transformation matrices:

$${}^{\text{map}}T_{\text{ee}} = {}^{\text{map}}T_{\text{odom}} \cdot {}^{\text{odom}}T_{\text{base}} \cdot {}^{\text{base}}T_{\text{arm_base}} \cdot {}^{\text{arm_base}}T_{\text{ee}}(\theta_1, \theta_2, \theta_3, \theta_4)$$

Docking accuracy is verified using LiDAR scan matching combined with simulated TF frame lookup.

3. Experimental Performance Results

Subsystem Pipeline	Baseline Unlinked	Proposed Integrated Nav2+MoveIt2	Gain
Docking to Grasp Latency	4.8 s	1.4 s	70.8% Faster
End-Effector Repeatability	±14.2 mm	±3.1 mm	78.1% Improvement
Pick-and-Place Success Rate	82.0%	96.5%	+14.5% Reliability

DIRECT RELEVANCE & SYNTHESIS FOR TEAM 3 PROJECT

Provides the exact software architecture for combining the TurtleBot3 mobile base navigation with the OpenManipulator-X 4-DOF arm in our simulation workcell.